

Chengtao Lin

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Experience & Team Projects

- Research Fellow** | *Artificial Intelligence Lab, UMass Boston* | [link](#) Jan 2024 – Present
- Work with the advisor on research direction and experiment design; own literature review and the analysis and preprocessing of research data to improve prediction quality.
 - Build reusable **Python experiment pipelines** (preprocessing, prototype training, automated result aggregation) and maintain documentation and version control so multi-month work stays traceable and reproducible.
- Tech Fellow** | *CodePath* | [link](#) Jul 2024 – Aug 2025
- Supported live online classes of **150+ students**, answering questions in real time; led 5–6 breakout groups (3–4 students each) through lab work, reviewing code and coaching students toward independent debugging.
 - Coordinated remotely with 3 managers and 15+ peers on task hand-offs, incident response, and student-feedback follow-through.
 - Nominated “**Best Tech Fellow**” by students multiple times, in class and in course feedback.
- Front-End Developer** | *YHack (Yale Hackathon)* | [link](#) · [demo](#) Spring 2026
- **Nominated for Best Prediction-Market Platform** among 600+ participants; built a working React client prototype from scratch with a team in **24 hours**.
 - Pulled linked market data through the Gamma API and **inferred market structure** from unstructured titles (strike ladders, expiry curves, mutually exclusive sets), then computed and ranked pricing dislocations.
 - Built **interactive Chart.js visualizations** that plot abstract no-arbitrage constraints as curves and envelopes.
- iOS Developer** | *HackUMass (UMass Amherst Hackathon)* | [link](#) Nov 2023
- Won **Best Mobile Hack** among 300+ participants; built Ree-See.it, a fully working iOS client prototype, from scratch with a team in **24 hours**.
 - Designed and implemented the login, receipt list, receipt detail, and receipt analytics screens, plus receipt categorization, map-based location, sign-up / auth / login, and on-device camera capture with optical character recognition (OCR); integrated the backend service to close the extraction loop.
- Backend Developer** | *Block by Block — Minecraft plugin tooling* Fall 2025
- Owned the **server-side API layer**: designed and implemented every endpoint (health check, player list, player detail, create / update / delete) on FastAPI, aligning the PostgreSQL schema with front-end needs.
 - Modeled **game runtime state** — player coordinates, health / hunger, playtime, inventory — into response structures and queries the front end and analytics could consume directly.
 - Integrated with the front end and controller, leaving hooks for downstream prediction / ML work.

Personal Projects

- Patient Education System — LM-judge pipeline** | *UMass BioNLP Lab · Python · LLM evaluation* | [link](#) Fall 2025
- Designed a **unified evaluation pipeline** spanning data audit, input extraction, batch submission, and result aggregation across Personalization, Doctor, and Imaging LM judges.
 - Compared GPT, Gemini, and MedGemma on radiology patient-education scenarios and iterated a dual-agent workflow and prompt versions; published on arXiv.
- iOS Client Collection** (PhotoScavengerHunt · BeReal Clone · Flixster) | *Swift* | [🐙 link](#) 2023 – 2024
- Built three complete iOS apps covering the core of **client engineering**: navigation structure, **app state and lifecycle**, **local persistence**, camera and photo-library permissions, Core Location / MapKit, networking and parsing.

Education

- University of Massachusetts Amherst** Amherst, MA
MS, Computer Science (GPA 3.8 / 4.0) 2025 – May 2026
- University of Massachusetts Boston** Boston, MA
BS, Computer Science 2019 – 2024

Skills

Languages: Python, Swift, JavaScript / TypeScript, SQL, Java, C
Client / App: SwiftUI, UIKit, MVVM, state management & lifecycle, local persistence, Core Location / MapKit, React
Tools & Systems: FastAPI · PostgreSQL · Docker · data modelling & schema design · data pipelines · LLM evaluation workflows
Collaboration: Git / GitHub, technical writing, remote collaboration